IMU orientation estimation under disturbances: a fair comparison on BROAD

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A gyroscope, an accelerometer and a magnetometer are enough to estimate 3D orientation, but magnets, steel and fast motion all corrupt the result. I compared Madgwick, Mahony and VQF with an error-state Kalman filter (ESKF) that I wrote, on the public BROAD benchmark. BROAD has 39 recordings with optical motion capture as ground truth, and 16 of them contain deliberate disturbances. The ESKF comes in two variants, with and without a check that skips the magnetometer when the field looks disturbed (called gated below). Every method was tuned the same way. Then I made the recorded data look like cheaper sensors and ran every method again without retuning.

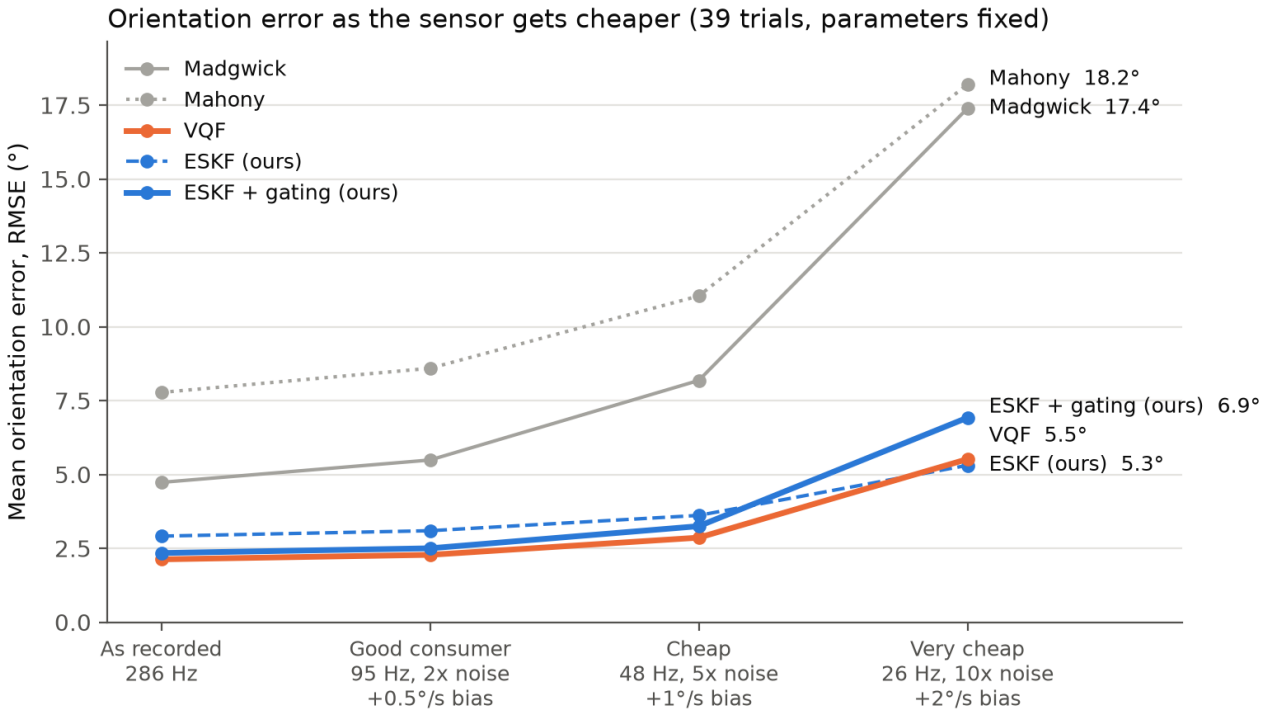


Figure 1. Mean orientation error over all 39 trials as the data are made to look like cheaper sensors: lower sampling rate, more noise and more gyroscope bias at the same time. The parameters stay as tuned on the original data. Solid blue is my gated ESKF, dashed blue the same filter without the check.

Main results

- VQF is the most accurate filter overall, with a mean error of 2.13° against 2.34° for my gated ESKF. On the 16 disturbed trials the two are level (2.66° and 2.62°). The gap is on the clean trials, and it is mostly heading error that builds up during long rotations.
- With a magnet attached 1 cm from the sensor (trial 32), the gated ESKF stays at 2.0°. The same filter without the check reaches 8.9°, and VQF 6.0°.
- An uncalibrated gyroscope bias of 5°/s breaks Madgwick, Mahony and VQF (38 to 48°). The ESKF stays at 2.5° because the bias is part of its state and has no fixed limit. In VQF the cause is a default bias limit of 2°/s, and raising the limit fixes it. So a filter's bias limit should be checked against the zero-rate offset in the gyroscope datasheet.
- VQF handles low sampling rates better. At 26 Hz it reaches 4.4°, the two ESKF variants 5.4° and 5.5°.

Code

Code, per-trial results and figures are at github.com/dorukyksl/robust-imu-orientation. The script `run_all.sh` downloads the data and recreates every number in this note.

Data and methods

Data. BROAD (Laidig et al., 2021) has 39 trials recorded at 286 Hz: 23 undisturbed and 16 with tapping, vibration, magnets near or on the sensor, or an office environment. The error is the RMSE of the orientation during the movement phase, split into heading and inclination as in the BROAD paper.

Filters. Madgwick and Mahony come from the *ahrs* package and VQF from the *vqf* package. I use VQF's real-time version, because its offline version looks at future samples. My ESKF estimates orientation and gyroscope bias (six error states). At rest it uses a zero angular rate update. In motion it corrects tilt with the accelerometer averaged over 3 s in the earth frame, an idea taken from VQF. The magnetometer corrects heading only. The gated variant skips the magnetometer while the field strength or dip angle is far from the Earth field (thresholds taken from VQF) and gives less weight to readings that fail a chi-square test.

Tuning. For each method a grid search over 12 to 42 parameter sets picked the one set with the lowest mean error over all 39 trials, as the BROAD paper does. This is in-sample. As a check I also scored each trial with the set that is best on the other 38 trials. VQF and the gated ESKF stay at 2.13° and 2.34°.

Cheaper sensors. White noise was added (2 to 20 times the recorded noise), blocks of samples were averaged down to 95, 48 and 26 Hz, and a constant gyroscope bias of 0.5 to 5°/s was added in a random direction.

Table 1. Mean orientation error after tuning (RMSE, degrees).

Method	Clean trials (23)	Disturbed trials (16)	All 39 trials
Madgwick	4.27	5.39	4.73
Mahony	6.78	9.21	7.77
VQF	**1.76**	2.66	**2.13**
ESKF (mine)	2.46	3.56	2.91
Gated ESKF (mine)	2.14	2.62	2.34

Table 2. Mean error with one sensor property made worse at a time, parameters as tuned (degrees).

Condition	Madgwick	Mahony	VQF	ESKF	Gated ESKF
As recorded	4.7	7.8	**2.1**	2.9	2.3
10x noise	6.2	7.9	5.2	3.4	3.4
20x noise	15.2	10.5	7.2	**3.8**	30.1
26 Hz	8.0	11.3	**4.4**	5.4	5.5
+5°/s gyroscope bias	47.8	38.6	41.6	3.4	**2.5**

What did not work, and the limits

- Learning the gyroscope bias from the accelerometer and magnetometer during motion made the estimate wander. On trial 01 the bias measured at rest is 0.47°/s, but the estimate moved between 0.2 and 0.6°/s during the motion and the heading drifted with it. Holding the bias fixed in motion once it has been measured at rest took the gated ESKF from 2.60° to 2.34°. The price is that a bias which changes during a long motion is only corrected at the next rest.
- The magnetic check in the gated ESKF uses fixed thresholds. At 20 times the recorded noise it rejects 93% of the readings on a clean trial and the error grows to 30°, while the ESKF without the check stays at 3.8°. Scaling the thresholds with the noise measured at rest would be the next step.
- Vibration gets past every check. The worst trial for both VQF and the gated ESKF is a phone vibration trial (6.9° and 6.6°).
- The structure of the ESKF was shaped by looking at a handful of trials, so those design choices are in-sample, even though its two noise parameters went through the same search as the other methods.

References

[1] D. Laidig, M. Caruso, A. Cereatti, T. Seel. BROAD - A Benchmark for Robust Inertial Orientation Estimation. *Data* 6(7), 72, 2021.

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[3] J. Solà. Quaternion kinematics for the error-state Kalman filter. arXiv:1711.02508, 2017.